SR.IIT *(MODEL-A)&ALL IIT **Date: 13-01-26**
Time: 3 Hrs **SPL GTM TEST-3** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A particle is projected with speed u at angle θ with horizontal from the bottom of an inclined plane of inclination $\alpha(\alpha < \theta)$. The range measured along the incline is maximum when :-

(A) $\theta = 45^\circ + \frac{\alpha}{2}$

(B) $\theta = 45^\circ + \alpha$

(C) $\theta = 90^\circ - \alpha$

(D) $\theta = 45^\circ$

2. A tunnel is dug along a chord of Earth at a distance $R/2$ from centre ($R =$ radius of Earth). Time period of oscillation of a particle performing SHM in this tunnel is :-

(A) $2\pi\sqrt{\left(\frac{R}{g}\right)}$ (B) $2\pi\sqrt{\left(\frac{2R}{3g}\right)}$ (C) $2\pi\sqrt{\left(\frac{R}{2g}\right)}$ (D) $2\pi\sqrt{\left(\frac{3R}{2g}\right)}$

3. One mole of an ideal diatomic gas undergoes a cyclic process ABCA where AB is isochoric ($T_A = 300\ K \rightarrow T_B = 600\ K$), BC is isobaric cooling to $300\ K$, CA is isothermal. Work done by gas in cycle is :-

(A) $300R\,\ell n2$ (B) $300R(1-\ell n2)$ (C) $-300R\,\ell n2$ (D) $300R(\ell n2-1)$

4. A battery of emf E and internal resistance r is connected to a variable resistance R. Maximum power dissipated in R occurs at $R = r$. If two such batteries are connected in parallel with same R , new resistance for maximum power is :-

(A) r (B) $\frac{r}{2}$ (C) 2 r (D) $\frac{r}{4}$

For More Join: @IITJEE_Advanced

5. In series LCR circuit, impedance at resonance is 100Ω. At frequency where $X_L = 2X_C$, current lags voltage by 30°. Power factor at this frequency is :-

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{1}{2}$ (D) $\frac{\sqrt{3}}{4}$

6. The dimensions of $\int \vec{E}.\vec{d\ell}$ (line integral of electric field) in the equation $\oint \vec{E}\cdot\vec{d\ell} = -\frac{d\phi_B}{dt}$ are

(A) $\left[MLT^{-2}\right]$ (B) $\left[ML^2\ T^{-3}\ A^{-1}\right]$ (C) $\left[ML^2\ T^{-2}\ A^{-1}\right]$ (D) $\left[ML^3\ T^{-3}\ A^{-1}\right]$

7. A convex lens of focal length f forms a real image of an object placed at 2f. The image is :-

(A) Virtual and magnified (B) Real and same size

(C) Real and magnified (D) Virtual and diminished

8. Magnetic susceptibility of diamagnetic material is :-

(A) Positive and large (B) Negative and small

(C) Zero (D) Positive and small

9. A particle moves under a force $F = -kx + \lambda x^3$. What is the nature of equilibrium at $x = 0$? (k & λ are positive constant)

(A) Stable (B) Unstable (C) Neutral (D) Depends on λ

10. A boat crosses a river of width d flowing with speed v_0. The boat's engine gives constant speed $3v_0$ relative to still water. The boat man always keeps the boat headed at 30^0 to the direction of river flow. The magnitude of drift when the boat reaches the opposite bank is :-

(A) $\frac{\sqrt{3}\ d}{2}$ (B) $\frac{d}{2}$ (C) $\frac{2\ d}{3}\left[1+\frac{3\sqrt{3}}{2}\right]$ (D) $\frac{2\ d}{3}\left[\frac{3\sqrt{3}}{2}-1\right]$

11. Two identical wires of cross-section area A and natural length L are connected end-to-end. One is steel (Y_S) and the other is copper $(Y_C = 1.2Y_S)$. A mass m attached to the copper end performs SHM when slightly displaced longitudinally. The effective Young's modulus is :-

(A) $\left(\frac{11}{5}\right)Y_S$ (B) $\left(\frac{6}{11}\right)Y_S$ (C) $\left(\frac{5}{6}\right)Y_S$ (D) $\left(\frac{12}{11}\right)Y_S$

12. A laminar stream is flowing vertically down from a tap of cross section area $1\ cm^2$. At a distance 10 cm below the tap, the cross section area of the stream has reduced to $\frac{1}{2}\ cm^2$. The volumetric flow rate of water from the tap must be about:

(A) 2.2 litre /min (B) 4.9 litre /min (C) 0.5 litre /min (D) 7.6 litre/min

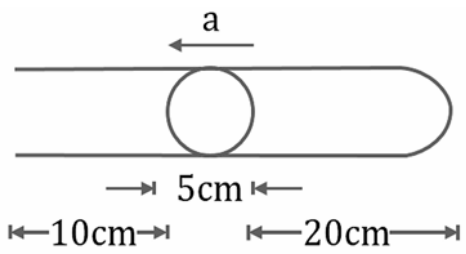
For More Join: @IITJEE_Advanced

13. A material has Poisson's ratio 0.5. If a uniform rod of this material suffers a longitudinal strain of 2×10^{-3}, the percentage increase in its volume is :

(A) 0% (B) 2% (C) 4% (D) 8%

14. A mercury pallet of length $5cm$ is trapped in a horizontal tube of small cross-section area. Length of tube enclosed by mercury pallet is $20cm$ when it is accelerated with $a = \frac{g}{2}$ in the direction shown. The length of tube enclosed by pallet if it is accelerated in opposite direction with same acceleration is :

(Atmospheric pressure is 72.5 cm of Hg)



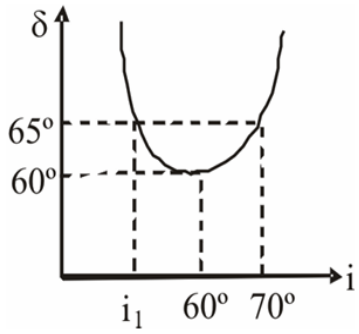
(A) 22 cm (B) 22.5 cm (C) 21 cm (D) 21.4 cm

15. Three identical point masses, each of mass 1 kg lie in the $x-y$ plane at points $(0,0)(0,0.2\ m)$ and $(0.2\ m, 0)$. The gravitational force on the mass at the origin is :-

(A) $1.67\times10^{-9}(-\hat{i}-\hat{j})N$ (B) $2.36\times10^{-9}(\hat{i}+\hat{j})N$

(C) $1.67\times10^{-9}(\hat{i}+\hat{j})N$ (D) $3.34\times10^{-10}(\hat{i}+\hat{j})N$

16. The angle of deviation (δ) vs angle of incidence (i) is plotted for a prism. Pick up the correct statement(s) :



(A) The angle of prism is 60^0

(B) The refractive index of the prism is $n=\sqrt{3}$

(C) For deviation to be 65^0 the angle of incidence $i_1=55^0$

(D) All the above

For More Join: @IITJEE_Advanced

17. In experiment to determine refractive index of glass slab using travelling microscope, following reading are obtained when microscope is focussed on :

(a) Cross mark without slab : 3.432 cm

(b) Cross mark with slab over it : 4.752 cm

(c) Lycopodium powder on slab : 7.152 cm

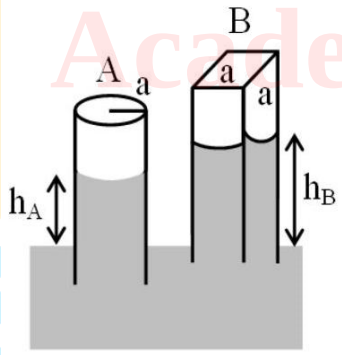
From above, refractive index of slab determined is :

(A) 1.45 (B) 1.49 (C) 1.53 (D) 1.55

18. In physics lab, a student performs experiment using resonance tube of diameter 5 cm filled with water. A tuning fork of frequency 420 Hz is sounded just above the tube. The zero of reference meter scale just coincides with top end of tube. The speed of sound at given temperature is $336\ m/s$. When first resonance occurs, the level of water level in the column is

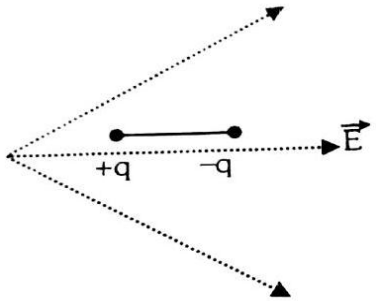
(A) 15.75 cm (B) 16.25 cm (C) 17.25 cm (D) 18.50 cm

19. Two capillary tubes A and B are of same length and material are partially dipped inside a liquid of density δ. Capillary A has circular cross-section of radius ' a '. While capillary B has square cross section of side a. The ratio of rise of liquid in capillary A and B is.



(A) $\pi : 1$ (B) $1 : \pi$ (C) $1 : 2$ (D) $2 : 1$

20. A dipole is placed in an electric field as shown. In which direction will it move?



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(A) towards the left as its potential energy will increase.

(B) towards the right as its potential energy will decrease.

(C) towards the left as its potential energy will decrease.

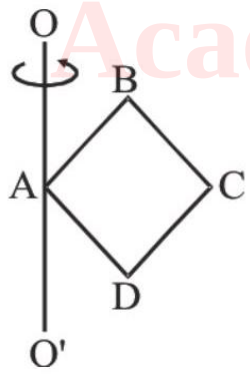
(D) towards the right as its potential energy will increase.

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. If a thermometer reads freezing point of water as 20^0C and boiling point as 150^0C, how much thermometer will read (in ^0C) when the actual temperature is 60^0C ?

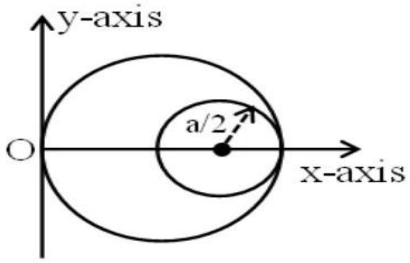
22. Four rods each of equal mass m and length ℓ forms a square ABCD as shown in figure. The moment of inertia of ABCD about the axis OO' passing through the point A lying in the plane of square is $\frac{\alpha}{3}m\ell^2$. The value of α is



23. A circular hole of radius $\left(\frac{a}{2}\right)$ is cut out of a circular disc of radius ' α ' as shown in figure.

The centre of mass of the remaining circular portion with respect to point ' O ' will be $\frac{\alpha}{6}a$.

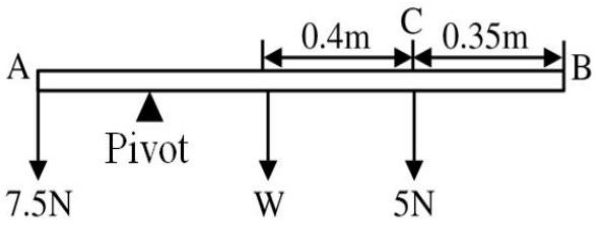
The value of α is



For More Join: @IITJEE_Advanced

24. In a damped oscillation of system, the time taken for amplitude to become half is 32 s , then the time (in sec) taken for mechanical energy of system to become half is :

25. A uniform rigid rod AB of length 1.5 m and 200 gm -wt is pivoted as shown. When a force of 7.5 N is applied at end A and 5 N at point C , calculate the distance of pivot point from end A (in cm) when rod is in equilibrium.



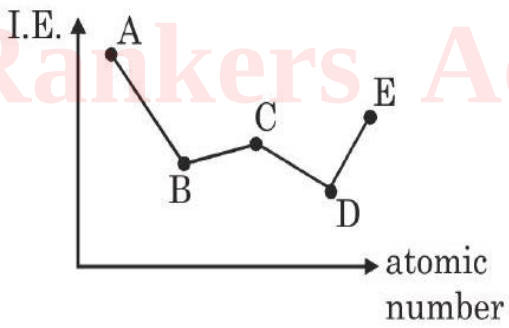
CHEMISTRY MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Carefully observe the following graph.



(A,B,C,D,E are elements of same group of periodic table)

The above graph is correct for which group of periodic table.

(A) Group : 1 (B) Group : 14 (C) Group : 13 (D) Group : 15

27. **Statement-I :** The shape of an orbital is defined as a surface of constant probability density that encloses some large fraction (say 90%) of the probability of finding the electron.

Statement-II : An s-orbital, for which ψ depends only on r and $|\psi|$ is constant on the surface of a sphere with center at the nucleus, has a spherical shape.

(A) Statement-I is correct, but Statement-II is incorrect.

(B) Statement-I is incorrect, but Statement-II is correct.

(C) Both statements are correct.

(D) Both statements are incorrect.

For More Join: @IITJEE_Advanced

28. For C(diamond), $C_{P,m} = 6.115\ J\ mol^{-1}\ K^{-1}$ at $25^{\circ}C$ and 1 bar. One carat diamond is 200 mg . C_P for a 10.0 -carat diamond is.

(A) $6.115\ J\ K^{-1}$ (B) $61.15\ J\ K^{-1}$ (C) $12.23\ J\ K^{-1}$ (D) $1.019\ J\ K^{-1}$

29. Which of the following order is CORRECT with respect to the given property?

(i) NO_2 and $C\ell F_3$ both are planar and polar molecules

(ii) $ZnCl_2 < CdCl_2 < HgCl_2$ (Covalent character)

(iii) $BeSO_4 < MgSO_4 < BaSO_4$ (Water solubility)

(iv) $NaHCO_3 < KHCO_3 < RbHCO_3$ (Thermal stability)

(v) $BeCl_2 > MgCl_2 > CaCl_2$ (Melting point)

(A) (i), (ii) & (iii) (B) (iv) & (v) (C) (i), (ii) & (iv) (D) (i), (ii) & (v)

30. For the reaction : $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$, $-\frac{d[H_2]}{dt} = 0.06 Ms^{-1}$ at a certain instant.

What is the value of $\frac{d[N_2]}{dt}$ at that instant?

(A) $0.18 Ms^{-1}$ (B) $-0.18 Ms^{-1}$ (C) $-0.02 Ms^{-1}$ (D) $0.02 Ms^{-1}$

31. In which of the following processes, the bond order has increased and paramagnetic character has changed to diamagnetic ?

(A) $N_2 \rightarrow N_2^+$ (B) $NO \rightarrow NO^+$ (C) $O_2 \rightarrow O_2^{2-}$ (D) $C_2 \rightarrow C_2^-$

32. Study the following statements :

(i) Λ_m for a strong electrolyte in water decreases as the electrolyte concentration increases.

(ii) Only at low concentration, conductivity for a strong electrolyte in water increases as the electrolyte concentration increases.

(iii) Conductivity for a weak electrolyte in water always increases as the electrolyte concentration increases.

The correct statement(s) is/are :

(A) Only (i) (B) Only (i) and (ii)

(C) Only (i) and (iii) (D) Only (ii) and (iii)

33. The vapours in equilibrium with an ideal solution of benzene and toluene at $20^{\circ}C$ has 30% benzene, by moles. The mole-fraction of toluene in the liquid solution at equilibrium, is.

(Given : At $20^{\circ}C$, vapour pressures of pure toluene and benzene are 25 torr and 75 torr, respectively)

(A) 0.70 (B) 0.875 (C) 0.4375 (D) 0.50

For More Join: @IITJEE_Advanced

34. Match the list:

	List-I (complex)		List-II (Properties)
P	$[CoF_6]^{3-}$	1	Diamagnetic and low spin complex
Q	$[Cr(C_2O_4)_3]^{3-}$	2	Paramagnetic and outer orbital complex
R	$[AuCl_4]^-$	3	Paramagnetic and inner orbital complex
S	$[Fe(CN)_6]^{3-}$	4	Paramagnetic and low spin complex

The correct option is

(A) $P \to 1; Q \to 2; R \to 3; S \to 4$ (B) $P \to 2; Q \to 3; R \to 1; S \to 4$

(C) $P \to 2; Q \to 1; R \to 3; S \to 4$ (D) $P \to 1; Q \to 3; R \to 2; S \to 4$

35. The dissociation constant of a weak acid, HA, is 2×10^{-6}.20ml of 0.5 M HA solution is diluted to 200 ml . Which of the following is the only incorrect information related with dilution?

(A) Degree of dissociation of HA increases.

(B) Moles of H^+ ion in solution increases.

(C) Concentration of A^- ion in solution decreases.

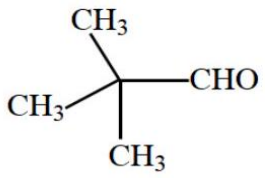
(D) Concentration of OH^- ion in solution decreases.

36. Which of the following is incorrect oxidation of alkane-

(A) $2CH_4 + O_2 \xrightarrow[100\ atm]{Cu/523\ K} 2CH_3OH$

(B) $CH_4 + O_2 \xrightarrow[\Delta]{M_{O_2}O_3} HCHO + H_2O$

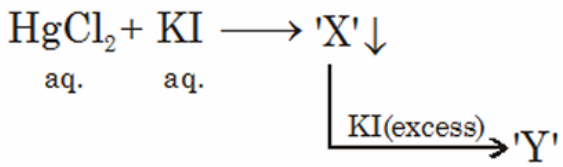
(C) $2CH_3 - CH_2 + 3O_2 \xrightarrow[\Delta]{(CH_3COO)Mn} 2CH_3COOH + 2H_2O$

(D) $(CH_3)_3CH \xrightarrow{KMnO_4}$ 

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37.



Select correct option :-

(A) ' Y ' is a mixture of black ppt. and water soluble complex

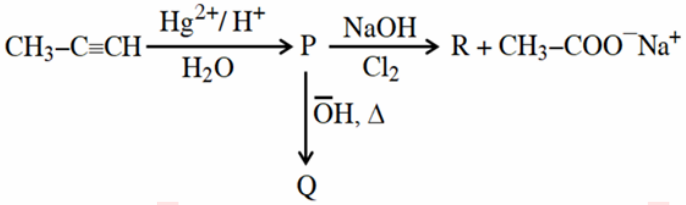
(B) ' X ' is green ppt.

(C) ' Y ' have sp^3 hybridization of Hg

(D) Neutral solution of ' Y ' is used to detect

NH_3 gas

38.



Which of the following is correct statement about P, Q, R .

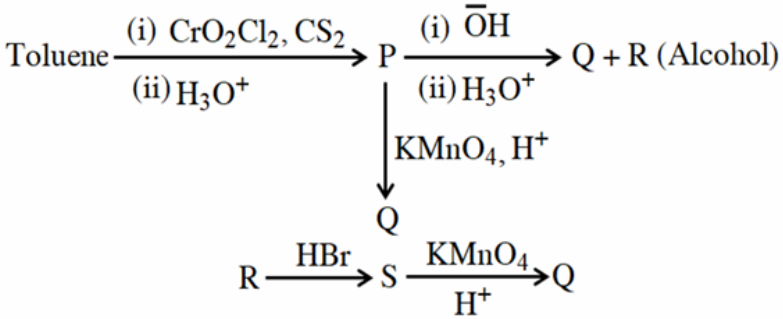
(A) Q is Mesityl oxide

(B) P gives colored precipitate with 2, 4-Dinitrophenyl-Hydrazine

(C) $R \xrightarrow[\Delta]{Ag}$ Acetylene

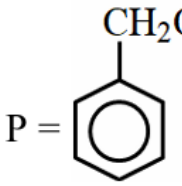
(D) All of these

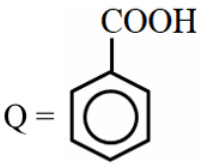
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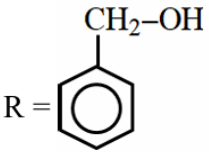


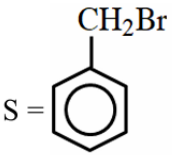
For More Join: @IITJEE_Advanced

Which of the following is incorrect according to above reactions

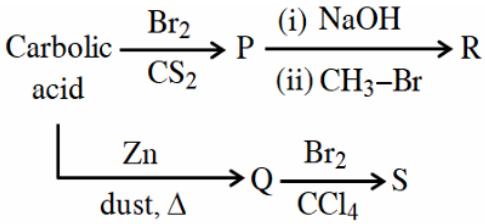
A) P = 

B) Q = 

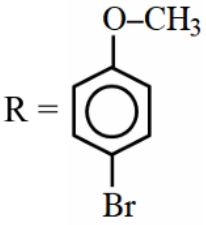
C) R = 

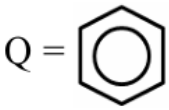
D) S = 

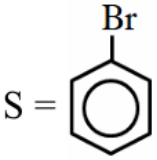
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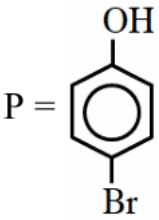


Incorrect major organic product for above reaction is-

(A) R = 

(B) Q = 

(C) S = 

(D) P = 

41. Which of the following statement is **INCORRECT** ?

(A) Np is one of the trans-uranium element

(B) Pm is the only synthetic radioactive lanthanide

(C) LnC_2 on hydrolysis gives C_2H_2

(D) Actinide contraction is result of poor shielding of 4 f -electrons.

42. Which of the following is wrong statement -

(A) Pentose sugar present in DNA is β-D-2-deoxyribose

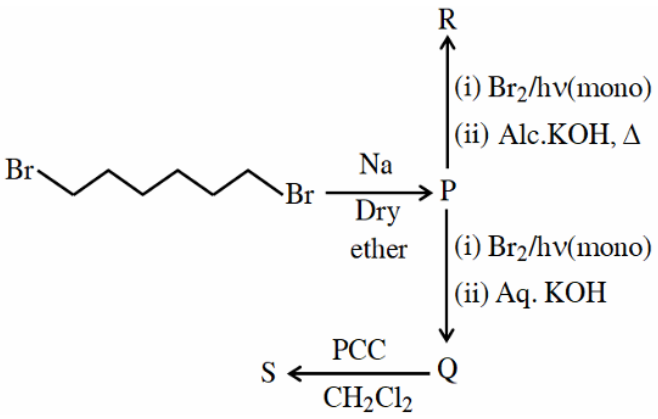
(B) Cellulose is the natural polymer where monomeric unit is β-D-glucopyranose

(C) In Denaturation of protein, primary structure of protein destroy

(D) Glucose (dextrose) does not give Schiff's test

For More Join: @IITJEE_Advanced

43.



Which of the following is correct statement according to above reaction-

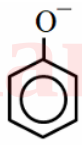
(A) R gives Baeyer's reagent test

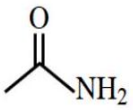
(B) $S \xrightarrow[HCl]{Zn(Hg)} P$

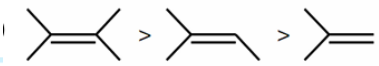
(C) $S \xrightarrow[H_2O]{NaBH_4} Q$

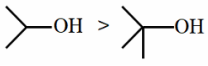
(D) All of these

44. Incorrect order according to given properties-

(A)  < $CH_3-\overset{\overset{O}{\|}}{C}-O^-$ (Stability of Anion)

(B)  < $Ph-NH_2$ (Basic strength)

(C)  (Heat of hydrogenation)

(D) $CH_3-CH_2-OH >$  > (Solubility in H_2O)

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45. Give the correct order of initials T or F for following statements. Use T if statement is true and F if it is false.

(i) Cu^{+} undergoes disproportionation to Cu and Cu^{2+} in aq. solution

(ii) Hg_2Cl_2 does not impart chromyl chloride test

(iii) $\left[Fe(H_2O)_5\,NO\right]^{2+}$ complex is highly unstable due to the presence of weak field ligand

(iv) Stability of ions : $Cr^{+3} > Fe^{+3} > Mn^{+3}$

(A) TFFT (B) TTFT (C) FTTF (D) TTFF

SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. Study the following processes applied independently, for a fixed amount of an ideal gas :

(i) Isothermal reversible expansion.

(ii) Isothermal irreversible compression.

(iii) Adiabatic reversible expansion.

(iv) Adiabatic irreversible expansion.

(v) Adiabatic irreversible compression.

(vi) Isobaric cooling.

(vii) Isochoric heating.

(viii) Adiabatic free expansion.

The number of processes in which $\Delta S_{sys} = +ve$, is.

47. At $25^{\circ}C$:

$E^{o}_{Ag^{+}(aq)|Ag} = 0.804\ V$

K_{sp} of $AgX(s) = 4\times10^{-14}$

and $\dfrac{2.303RT}{F} = 0.06\ V$

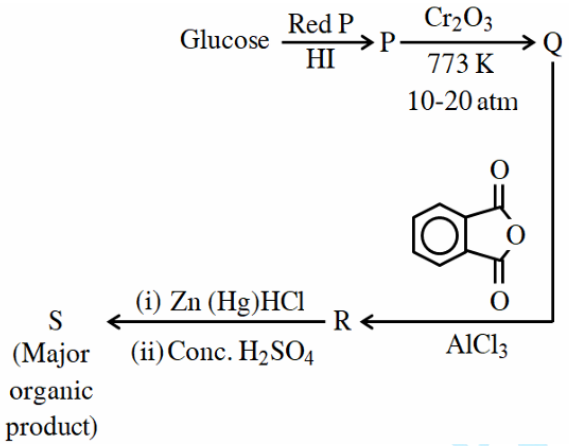
The value of $E^{o}_{X^{-}(aq)|AgX(s)|Ag(s)}$ at $25^{\circ}C$ is (x V).

The value of ' x ' is ($\log 2 = 0.30$)

(Answer to the nearest integer)

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48.



Degree of unsaturation of major product S.

49. Among the following cations(s) :

$Hg^{2+}, Mg^{2+}, Cu^{2+}, Ag^{+}, Pb^{2+}$

(i) Number of cation(s) gives(s) yellow ppt with Na_2SO_4 is $= (X)$

(ii) Number of cation(s) gives(s) yellow ppt with KI is $= (Y)$

(iii) Number of cation(s) gives(s) white ppt with HCl is $= (Z)$

Find the value of $(X + Y + Z)$

50. Calculate CFSE for spin paired octahedral complex of M^{+2} metal cation, where M belongs to group which are having highest melting point out of all transition elements. [answer should be given in deciquanta unit (dq) and only numerical value, $(1\Delta_o = 10\,dq)$]

MATHEMATICS

MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. If $\vec{V}_1 = \hat{i} + \hat{j} + \hat{k}, \vec{V}_2 = a\hat{i} + b\hat{j} + c\hat{k}$

where $a, b, c \in \{-2, -1, 0, 1, 2\}$, then the number of non-zero vectors $\vec{V}_2$ which are perpendicular to $\vec{V}_1$ are

(A) 18 (B) 24 (C) 21 (D) 12

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52. Let L be the line passing through the point $P(1,2)$ such that its intercepted segment between the co-ordinate axes is bisected at P . If L_1 is the line perpendicular to L and passing through the point $(-2,1)$ then the point of intersection of L and L_1 is

(A) $\left(\frac{3}{5},\frac{23}{10}\right)$ (B) $\left(\frac{4}{5},\frac{12}{5}\right)$ (C) $\left(\frac{11}{20},\frac{29}{10}\right)$ (D) $\left(\frac{3}{10},\frac{17}{5}\right)$

53. **Assertion (A)** : Number of terms in expansion of $(a+b+c)^{20}$ is 231 .

and

Reason (R) : Total number of non-negative integral solutions of equation $x+y+z=n$ is ${}^{n+2}C_3$.

(A) (A) is True, (**R**) is True ; (**R**) is a correct explanation for (A).

(B) (A) is True, (R) is True ; (R) is not a correct explanation for (A).

(C) (A) is True, (R) is False.

(D) (A) is False, (R) is True.

54. Suppose $A_1, A_2,\ldots, A_{30}$ are thirty sets each having 5 elements and $B_1, B_2,\ldots, B_n$ are n sets each with 3 elements, let $\bigcup_{i=1}^{30} A_i = \bigcup_{i=1}^{n} B_j = S$ and each element of S belongs to exactly 10 of the A_i's and exactly 9 of the B'_js, then ' n ' is equal to

(A) 40 (B) 54 (C) 45 (D) 27

55. Let $A=\{1,2,3,4,5,6,7\}$. The number of surjective function defined from A to A such that $f(i)=i$ for atleast 4 values of i from $i=1,2,\ldots,7$, is

(A) 7! (B) 92 (C) 126 (D) 407

56. The value of the expression $\frac{\sin^2\frac{2\pi}{7}}{\sin^2\frac{\pi}{7}}+\frac{\sin^2\frac{4\pi}{7}}{\sin^2\frac{2\pi}{7}}+\frac{\sin^2\frac{\pi}{7}}{\sin^2\frac{4\pi}{7}}$ is equal to

(A) 3 (B) 4 (C) 5 (D) 6

57. The set of values of ' a ' for which all extremum of function $f(x)=x^3-3ax^2+3(a^2-1)x+1$ lie in the interval $(-2,4)$ is

(A) $(3,4)$ (B) $(-1,3)$ (C) $(-3,-1)$ (D) $(-2,-1)$

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58. If $f(x)=\begin{cases}-e^{-x}+k, & x\leqslant 0\\ e^{x}+1, & 0<x<1\\ ex^{2}+\lambda, & x\geqslant 1\end{cases}$,one-one and monotonically increasing for all $x\in R$,

then difference of maximum value of k and minimum value of λ is

(A) 0 (B) 1 (C) 2 (D) 3

59.

List-I		List-II	
(A)	The vector $(\vec{a}+3\vec{b})$ is perpendicular to $(7\vec{a}-5\vec{b})$ and vector $(\vec{a}-4\vec{b})$ is perpendicular to $(7\vec{a}-2\vec{b})$. Then angle between $\vec{a}$ & $\vec{b}$ is	(I)	30^0
(B)	For non-zero two vectors $\vec{a}$ and $\vec{b}$, if $\lvert\vec{a}\cdot\vec{b}\rvert=\lvert\vec{a}\times\vec{b}\rvert$, then the angle between $\vec{a}$ & $\vec{b}$ is	(II)	45^0
(C)	Angle between the line $\frac{x+3}{4}=\frac{y-4}{-2}=\frac{z+5}{-1}$ and plane $4x-2y-z=1$ is	(III)	60^0
(D)	If a,b,c are different real numbers and $a\hat{i}+b\hat{j}+c\hat{k}$; $b\hat{i}+c\hat{j}+a\hat{k}$ & $c\hat{i}+a\hat{j}+b\hat{k}$ are position vectors of three non-collinear points A, B & C then angle between $\overrightarrow{AB}$ and $\overrightarrow{AC}$ is	(IV)	90^0

(A) (A) → (III), (B) → (II), (C) → (IV), (D) → (III)

(B) (A) → (III), (B) → (IV), (C) → (II), (D) → (I)

(C) (A) → (II), (B) → (III), (C) → (IV), (D) → (I)

(D) (A) → (I), (B) → (III), (C) → (IV), (D) → (III)

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60. If $\alpha,\beta,\gamma \in R^+$ such that $\alpha+\beta+\gamma=\sqrt{3}$, then minimum value of $\left(\frac{\pi}{\tan^{-1}\alpha+\tan^{-1}\beta+\tan^{-1}\gamma}\right)$ is

(A) 2 (B) 3 (C) 5 (D) 6

61. Let $P\equiv(1,7,\sqrt{2})$ be a point and line L is $2\sqrt{2}(x-1)=y-2,z=0$. If PQ is the distance of the plane $\sqrt{2}x+y-z=1$ from point P measured along a line inclined at an angle 45° with the line L and is minimum, then value of PQ equals

(A) 3 (B) 4 (C) 5 (D) 6

62. Let $A(z_1),B(z_2)$ and $C(z_3)$ lie on the circle $|z-i|=1$ and satisfy the equation $3z_1+i=2z_3+2z_2$. If D is the centre of the circle $|z-i|=1$, then area of quadrilateral ABCD equals

(A) $\frac{\sqrt{3}}{4}$ (B) $\frac{\sqrt{5}}{4}$ (C) $\frac{\sqrt{11}}{4}$ (D) $\frac{\sqrt{7}}{4}$

63. If 3 whole numbers are randomly selected and multiplied then the chance that the unit place in their product is 0 is λ, then the value of 1000λ is

(A) 407 (B) 427 (C) 513 (D) 527

64. Let a and b are two positive real numbers such that $a^2+b=2$, then the maximum value of term independent of x in the expansion of $\left(ax^{1/6}+bx^{-1/3}\right)^9$ is

(A) 80 (B) 70 (C) 84 (D) 75

65. Statement-I : If algebraic sum of length of perpendiculars drawn from points $A(4,2),B(2,4)$ and $C(0,0)$ to variable line is zero, then this line passes through $(2,2)$.

and

Statement-II : If algebraic sum of length of perpendiculars drawn from three non-collinear points $A(x_1,y_1),B(x_2,y_2)$ and $C(x_3,y_3)$ to a variable line is zero, then this line passes through centroid of $\triangle ABC$.

(A) Both Statement-I and Statement-II is true

(B) Both Statement-I and Statement-II is false

(C) Statement-I is true but Statement-II is false

(D) Statement-I is false but Statement-II is true

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66. A function $f:R\to R$ satisfies the equation $f(x)f(y)-f(xy)=x+y,\forall x,y\in R$ and $f(1)>0$, then $f(2)f^{-1}(2)$ equal to

(A) 1 (B) 3 (C) 4 (D) 6

67. If $\sum_{r=1}^{19}\frac{3(r+1)r+1}{(r+1)(r^5+2r^4+r^3)}=K$ then 1000 K is

(A) 999.875 (B) 860.505 (C) 765.155 (D) 699.185

68. $\int e^{\sin x}\left(1+\ell n(\sin x)^{\sin^2 x}\right)\cot x\,\operatorname{cosec} x\,dx$ is equal to (Where C is constant integration)

(A) $e^{\sin x}\left(\sin x+\ell n(\sin x)\right)+C$ (B) $e^{\sin x}\left(\ln(\sin x)-\operatorname{cosec} x\right)+C$

(C) $e^{\sin x}\left(\ell n(\sin x)-\sin x\right)+C$ (D) $e^{\sin x}\left(\ell n(\sin x)+\operatorname{cosec} x\right)+C$

69. Let a function 'f' satisfies $f(-x)=f(x)$ and $f(3+x)=f(1-x)\forall x\in R$ and

$f(x)=\begin{cases}x & ;\ 0\leqslant x\leqslant 1\\ 1-2x & ;\ 1<x\leqslant 2\end{cases}$. The number of points where function $f(x)$ is discontinuous in $[0,100]$ is equal to D and value of $\int_0^{100}f(x)dx$ is equal to I, then the value of $(2D+I)$ is equal to

(A) 32 (B) 25 (C) 30 (D) 22

70. Let slope of a tangent at point P on the ellipse $\frac{x^2}{81}+\frac{y^2}{9}=1$ is m such that normal at P is at a maximum distance from origin, then $|3m|$ is equal to

(A) $2\sqrt{3}$ (B) $3\sqrt{3}$ (C) $\sqrt{3}$ (D) $4\sqrt{3}$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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71. Let $\omega=e^{i\left(\frac{2\pi}{3}\right)}$ and a,b,c,x,y,z be non-zero complex number such that

$a+b+c=x$,

$a+b\omega+c\omega^2=y$,

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$a+b\omega^2+c\omega=z$.

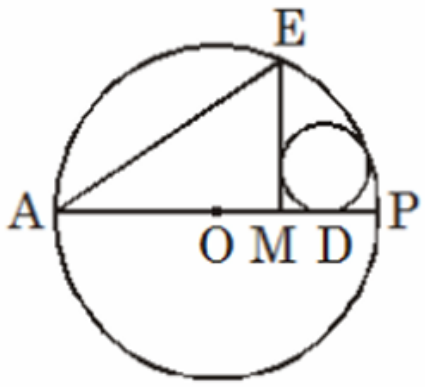
Then the value of $\dfrac{|x|^2+|y|^2+|z|^2}{|a|^2+|b|^2+|c|^2}$, is equal to

72. Consider the set of all triangles OPQ where O is the origin and P and Q are distinct points in the plane with non-negative integral coordinates (x, y) such that $5x+y=99$. Number of such distinct triangles whose area is an integer is

73. Using all of $0,0,0,1,1,1,-1,-1,-1$, a set 'S' of all 3×3 matrices formed. If number of symmetric matrices in S is K and number of matrices having trace zero in S is P, then the value of $\left(\dfrac{P}{10}+K\right)$ is

74. Let a and b be real numbers such that $9a^2+8ab+7b^2\le6$, then the maximum possible value of the expression $(7a+5b+12ab)$ is

75. Let 'O' be centre of circle as shown in figure and AE be a chord from A the diameter AP is drawn and M is foot of perpendicular drawn from E on AP. A circle is drawn such that it touches EM, AP at D and circle whose diameter is AP, then the value of $\dfrac{AE}{AD}$ is



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